

**Effects of a 250-Milliliter Red Bull Energy Drink Versus Water on Working Memory
Performance in Teenagers**

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Abstract

This study examined the acute effects of a 250-milliliter Red Bull energy drink compared to plain water on working memory performance in adolescents aged 13-17. Using a randomised, single-blind, within-subjects crossover design with an eight-participant sample, participants completed two cognitive assessments—the Word Span Backward (WSB) and a computerised 2-back task—following consumption of each beverage, separated by at least a seven-day washout period. Paired-samples t-tests revealed no statistically significant difference between Red Bull and water conditions for either measure: WSB accuracy, $t(7) = 1.369$, $p = .213$, $d = 0.48$; 2-back accuracy, $t(7) = 0.391$, $p = .708$, $d = 0.14$. Although descriptive trends favoured Red Bull on the WSB task, these differences did not reach significance. Findings are consistent with mixed adult literature and likely reflect limited statistical power due to small sample size. Results do not support the hypothesis that a single 250-milliliter serving of Red Bull produces meaningful working memory benefits in adolescents relative to water. Implications for youth health and future research directions are discussed.

Keywords: working memory, energy drinks, Red Bull, adolescents, caffeine, cognitive performance

Effects of a 250-Milliliter Red Bull Energy Drink Versus Water on Working Memory Performance in Teenagers

Energy drink consumption has become very common among adolescents, who often use beverages such as Red Bull to enhance alertness, improve concentration, or manage academic and extracurricular demands. Red Bull contains caffeine, taurine, glucose, and B-vitamins, which are marketed as cognitive enhancers. However, there is still limited research on how these substances actually affect teenagers. This gap is particularly noteworthy because adolescence is a period of rapid neurodevelopment during which stimulants may influence cognitive systems differently than in adults. Understanding whether Red Bull meaningfully alters working memory—the cognitive system responsible for temporarily storing and manipulating information—is therefore essential for educators, parents, and policymakers.

Existing research in adults suggests that energy drinks have near-instant effects on cognitive performance. According to Wesnes et al. (2017), when university students were given a 250-milliliter serving of Red Bull, their reaction time and certain working memory measures improved. Nonetheless, these findings cannot be assumed to generalise to teenagers, whose developing brains are more sensitive to caffeine and other stimulants (Temple, 2009). At the same time, research on hydration demonstrates that even plain water can improve cognitive performance in children (Khan et al., 2019), raising the question of whether Red Bull's effects exceed those of a neutral, non-stimulant beverage.

My interest in this topic emerged after observing peers frequently consuming energy drinks before exams, sports, and study sessions. Many of them believe that energy drinks enhance cognitive performance, yet the scientific evidence for adolescents remains unclear. This curiosity led to the development of the research question: To what extent does consuming a

250-milliliter Red Bull energy drink, compared to plain water, influence working memory performance in teenagers?

Based on prior findings in adult samples and ingredient-specific research, my hypothesis is that adolescents will demonstrate modest improvements in working memory performance after consuming Red Bull compared to water, particularly on tasks requiring rapid information updating. However, given concerns about prefrontal cortex maturation and heightened sensitivity to stimulants during adolescence, and mixed findings in the literature, the size of the effect was expected to be small. This study aims to address the lack of adolescent-focused experimental research and provide evidence relevant to youth health and cognitive performance.

Literature Review

Working Memory In Adolescence

Working memory is a core cognitive system responsible for temporarily storing and manipulating information during tasks such as reasoning, comprehension, and decision-making. Baddeley (2003) conceptualises working memory as a multi-component system consisting of verbal, spatial, and executive subsystems that interact to support higher-order cognition. Working memory is an underlying factor in academic performance and everyday problem-solving, which is why it is frequently used to evaluate the cognitive effects of stimulants, hydration, and nutritional interventions.

Adolescence is a developmental period characterised by ongoing maturation of the prefrontal cortex, which plays a central role in working memory. This developmental context is important because substances that influence arousal, such as caffeine, may affect adolescents differently than adults. Temple (2009) notes that adolescents may experience both short-term

cognitive benefits and heightened risks, including increasing distractibility and anxiety, which can counteract potential gains. Understanding how working memory functions in adolescence provides a foundation for evaluating whether energy drinks influence cognitive performance in this age group.

Energy Drinks and Cognitive Performance: Evidence from Adult Studies

A substantial body of research has examined the acute cognitive effects of energy drinks in adults. Wesnes et al. (2017) conducted a randomised, double-blind, placebo-controlled crossover trial and found that consuming a 250-milliliter Red Bull improved reaction time and certain working memory measures. These findings suggest that Red Bull's combination of caffeine, glucose, and taurine may enhance cognitive processing speed and attentional control. Goel et al. (2014) similarly reported that Red Bull consumption led to faster auditory reaction times, indicating enhanced neural responsiveness.

However, not all adult studies report positive effects. Moore (2014) found no significant differences in reaction time or executive functioning between participants who consumed Red Bull, a placebo, or no beverage. Eckerson et al. (2013) also reported null effects on physical performance after consuming sugar-free Red Bull, suggesting that stimulant-related benefits may not generalise across all performance measures. These mixed findings highlight the complexity of energy drink effects and underscore the need to examine specific cognitive outcomes, such as working memory, rather than assuming broad enhancement. A critical consideration when interpreting these adult findings for the current study is that adolescents differ from adults in core neurological ways. The prefrontal cortex—the part of the brain most directly involved in working memory—does not reach full maturity until the mid-twenties (Temple, 2009). During adolescence, adenosine receptors, which are the primary targets of caffeine's stimulant action, are undergoing

changes in development that could potentially alter sensitivity and magnitude of response. This means that even if caffeine reliably produces working memory benefits in adults through altering neural activity, the same mechanism may produce attenuated, absent, or unexpectedly negative effects in adolescents whose receptor systems are still developing.

Cognitive Effects of Energy Drink Ingredients

Because energy drinks contain multiple active ingredients, researchers have attempted to isolate the cognitive effects of individual components. Giles et al. (2012) investigated caffeine, taurine, and glucose both separately and in combination. Their findings revealed that caffeine primarily improved reaction time and alertness, glucose enhanced memory and accuracy, and taurine produced variable effects. Importantly, the combination of ingredients produced cognitive outcomes that differed from any single component alone, suggesting synergistic interactions.

This complexity helps explain why Red Bull's effects cannot be attributed solely to caffeine and why ingredient-specific research is insufficient for understanding how the full beverage influences working memory. Additionally, studies such as Bawazir (2017) show that chronic consumption of energy drink ingredients may negatively affect neural structures in animal models, raising questions about long-term safety even if short-term cognitive effects appear beneficial.

Hydration and Cognitive Performance in Youth

Hydration research provides an essential comparison point for evaluating Red Bull's effects. Khan et al. (2019) conducted a randomised crossover trial with children and found that increased water intake improved cognitive flexibility and reduced working memory cost. These findings illustrate that even plain water can enhance cognitive performance in younger populations. Because Red Bull contains water in addition to stimulants, hydration alone may

contribute to any observed cognitive changes. Therefore, using water as a control condition is necessary for isolating the effects of Red Bull's active ingredients.

Energy Drink Consumption in Adolescent Populations

Understanding why adolescents consume energy drinks provides context for interpreting cognitive outcomes. Malinauskas et al. (2007) found that young adults frequently consume energy drinks for perceived cognitive benefits such as increased alertness and concentration. Raj et al. (2018) similarly reported that many students believe caffeine improves focus, despite limited awareness of potential side effects. These findings suggest that expectancy effects may influence perceived cognitive enhancement.

Public health organisations express concern about these consumption patterns. The American Academy of Child and Adolescent Psychiatry (AACAP, n.d.) advises limiting caffeine intake to 100 milligrams per day for adolescents and discourages energy drink consumption altogether. The National Center for Complementary and Integrative Health (2018) warns that energy drinks may cause increased heart rate, anxiety, and sleep disturbances. Elitok et al. (2015) provide physiological evidence for these concerns, demonstrating that Red Bull acutely alters cardiac electrophysiology in young adults. These findings underscore the importance of studying energy drink effects specifically in adolescents.

Synthesis of Literature and Identified Research Gap

Across the literature, several themes emerged. First, adult studies convey that Red Bull can acutely influence cognitive performance, but these findings cannot be assumed to generalise to adolescents. Second, ingredient-specific research highlights the complexity of energy drink formulations, suggesting that cognitive effects arise from interactions among caffeine, glucose, and taurine. Third, hydration research shows that water alone can improve cognition in children,

making it a necessary control condition. Fourth, adolescents exhibit unique physiological and neurological responses to caffeine, raising concerns about safety and generalisability. Finally, no existing studies directly compare Red Bull and water in adolescent working memory performance using an experimental crossover design.

Therefore, a clear methodological and population gap exists: there is limited experimental research on the acute cognitive effects of a standard 250-milliliter Red Bull energy drink in adolescents, particularly in comparison to water and specifically in the domain of working memory. Wesnes et al. (2017) used adult university students, Giles et al. (2012) studied isolated ingredients rather than the full beverage in adolescents, and Khan et al. (2019) examined hydration without any stimulant condition. None of these studies, however, individually or collectively, provides evidence about whether a complete Red Bull serving produces working memory changes in teenagers beyond what water alone provides. This study directly addresses that intersection.

Based on the literature, the present study hypothesises that adolescents will show modest improvements in working memory performance after consuming Red Bull compared to water, though the effect is expected to be small due to developmental vulnerability and mixed findings in prior research.

Method

This study used a randomised, single-blind, within-subjects crossover experiment to examine the effects of the consumption of a 250-milliliter Red Bull compared to plain water on adolescent working memory performance. A crossover design was selected because it allows each participant to serve as their own control, which reduces individual variability and increases statistical power in small samples. This structure is common in cognitive-nutrition research and

has been proven to detect short-term cognitive changes effectively (e.g., Wesnes et al., 2017; Khan et al., 2019).

A single-blind procedure was implemented in which the researcher administering and scoring the cognitive tasks remained unaware of beverage assignment. Full double-blinding was unrealistic because of the distinct taste between Red Bull and water, making it obvious to the participant which drink is which. Blinding still minimised expectancy bias during data collection.

Participants

A total of eight healthy adolescents aged 13-17 participated in the study. The target sample size was 16-20 participants; however, the final sample was reduced due to incomplete parental consent forms, failed policy compliances (such as consuming caffeine within the session window), and scheduling conflicts that prevented potential participants from completing both sessions within the collection window. Participants were recruited through social networks (primarily Instagram), classroom recruitment, and social media flyers. Recruitment flyers were distributed via Instagram posts and classroom announcements, each directing interested individuals to an online interest form via QR code or link (see Appendix A). Interested participants completed the form (see Appendix B), which collected demographic information and screened for basic eligibility. Eligible participants were then sent the combined parental consent and assent form (see Appendix C), which was required to be completed before any session was scheduled. Eligibility required general good health, no cardiovascular conditions, no daily energy drink or caffeine consumption, and no diagnosed attention-related disorders. Exclusion criteria included caffeine sensitivity or allergy, medications that interact with caffeine, anxiety or sleep disorders, irregular caffeine consumption patterns, and cardiovascular or other health conditions.

Parents or guardians completed a health-screening questionnaire, and written parental consent and participant assent were obtained prior to participation.

Materials

The experimental condition consisted of a 250-milliliter serving of Red Bull (original flavour; approximately 80 mg of caffeine). The control condition was a 250-milliliter serving of plain bottled water. Both beverages were served in identical opaque cups to maintain researcher blinding.

Two validated working memory assessments were used. First, the Word Span Backward (WSB) test measured verbal working memory by requiring participants to repeat sequences of spoken words in reverse order (see Appendix E). The second test, a computerised 2-back task compiled on PsyToolKit (Stoet, 2010, 2017), measured spatial working memory and updating by requiring participants to identify when the current letter matched the one presented two trials earlier (see Appendix F). These two instruments were selected not only for their individual validity but because they provide complementary coverage of working memory; the WSB targets the phonological loop and executive control involved in verbal recall, while the 2-back task targets visual-spatial updating and central executive functioning. Using both instruments reduces the risk that any beverage effect is task-specific and allows for a more comprehensive assessment of working memory as a multi-component system (Baddeley, 2003). These tasks were selected because they are brief (5 and 10 minutes, respectively), reliable, and sensitive to short-term cognitive changes. The 2-back task has been used in prior energy-drink research (Wesnes et al., 2017). Additional materials included a digital timer, a laptop with stable WiFi, and standardised questionnaires for baseline screening, post-session side effects, and field-note documentation (see Appendices D, G, and H).

Procedure

Each participant completed two sessions of testing, one for each beverage. Sessions were separated by a minimum seven-day washout period to eliminate carryover effects and were scheduled between 8:00 a.m. and 3:00 p.m. to control for circadian influences on cognition.

Participants were instructed to abstain from caffeine for 24 hours prior to each session, obtain 7–9 hours of sleep the night before, eat a light meal at least one hour prior, and avoid strenuous exercise on testing days. Compliance was confirmed using the Pre-Session Health Screening and Baseline Questionnaire.

Upon arrival, participants completed the baseline questionnaire assessing alertness, mood, sleep quality, and adherence to pre-session requirements (see Appendix D). A third-party preparer, following a randomised order schedule, poured the assigned beverage into a coded opaque cup. Participants consumed the beverage within five minutes while being observed.

A 30-minute absorption period followed, consistent with pharmacokinetic evidence that caffeine reaches peak plasma concentration approximately 30–60 minutes post-ingestion (Temple, 2009), during which participants engaged in quiet, non-cognitively demanding activities. This timing aligns with evidence that caffeine typically reaches peak plasma concentration within 30–60 minutes. Participants then completed the WSB test (approximately 5 minutes) (see Appendix E) and the computerised 2-back task (approximately 10 minutes) (see Appendix F) during the expected peak window of cognitive effect. Participants then completed the Post-Session Side Effects and Cognitive State Questionnaire (see Appendix G).

While individual components of each session varied in duration, the overall session length was approximately 45–60 minutes.

Randomisation and Blinding

Participants were randomly assigned to one of two counterbalanced order sequences: Red Bull followed by water (Order A; $n = 4$) or water followed by Red Bull (Order B; $n = 4$). A third-party preparer maintained the randomisation key and prepared all beverages. The researcher remained blind to beverage assignment until all data was collected and recorded.

Data Collection

For each participant and each session, the researcher recorded Word Span Backward accuracy and 2-back accuracy. Baseline alertness and mood ratings, post-session side effects, and field-note observations (see Appendix H) were also documented. All data required for analysis was coded by participant ID and entered into a secure spreadsheet. Identifying information was stored separately from performance data. The completed anonymised dataset is available in Appendix I.

Data Analysis

Paired-samples t-tests were used to compare Red Bull and water conditions for Word Span Backward accuracy and 2-back accuracy. Effect sizes (Cohen's d) were calculated to assess the magnitude of differences. If practice effects were detected, such as consistent improvement in Session 2 regardless of beverage, a repeated-measures ANOVA was planned to separate beverage effects from order effects, although this analysis was not required. Inspection of session means indicated no consistent practice effect (Session 1 $M = 6.63$, Session 2 $M = 7.25$ on WSB, calculated by collapsing across beverage conditions within each session order), and the counterbalanced design mitigated order confounds. Post-session questionnaire responses were summarised to contextualise cognitive findings.

Ethical Considerations

The study followed ethical guidelines for research with minors. Parental consent and participant assent were required. The caffeine dose (80 mg) remained within the American Academy of Child and Adolescent Psychiatry (AACAP) recommended daily limit for adolescents. Participants could withdraw at any time without penalty. All data was anonymised, stored securely, and used solely for research purposes.

Results

Table 1 *Individual Participant Scores by Beverage Condition*

Participant	Order	WSB (Red Bull)	WSB (Water)	2-Back RB (%)	2-Back Water (%)
1	A	7	9	80	68
2	A	7	6	60	68
3	A	7	4	48	12
4	B	9	7	52	60
5	A	8	9	85	80
6	B	8	7	85	100
7	B	6	6	52	60
8	B	7	4	72	68
M (SD)	—	7.38 (0.92)	6.50 (1.93)	66.75 (15.59)	64.50 (24.88)

Table 1 presents the raw scores for each participant across both beverage conditions. Descriptive statistics and inferential test results are summarised in Table 2.

Note. WSB = Word Span Backward (number of sequences correctly recalled in reverse order; maximum possible score = 14). 2-Back RB = 2-back accuracy (%) under Red Bull condition. Order A = Red Bull first; Order B = Water first.

Word Span Backward

Mean WSB accuracy was higher in the Red Bull condition ($M = 7.38$, $SD = 0.92$) than in the water condition ($M = 6.50$, $SD = 1.93$). A paired-samples t -test revealed that this difference was not statistically significant, $t(7) = 1.369$, $p = .213$, $d = 0.48$. The effect size of $d = 0.48$ falls in the small-to-medium range (Cohen 2013), suggesting a practically meaningful but statistically underpowered trend. Inspection of individual scores revealed notable variability: two participants (IDs 1 and 5) scored higher on water, while three (IDs 3, 6, and 8) showed markedly higher WSB scores under Red Bull.

2-Back Task

Mean 2-back accuracy was slightly higher in the Red Bull condition ($M = 66.75\%$, $SD = 15.59$) than in the water condition ($M = 64.50\%$, $SD = 24.88$). A paired-samples t -test indicated that this difference was not statistically significant, $t(7) = 0.391$, $p = .708$, $d = 0.14$. The near-zero effect size suggests negligible difference between conditions on this measure. High variability in the water condition ($SD = 24.88$) is notable and likely contributed to reduced statistical power.

Table 2 *Paired-Samples t-Test Results for Working Memory Measures*

Measure	Red Bull M (SD)	Water M (SD)	t(7)	p	Cohen's d
Word Span Backward	7.38 (0.92)	6.50 (1.93)	1.369	.213	0.48
2-Back Accuracy (%)	66.75 (15.59)	64.50 (24.88)	0.391	.708	0.14

Note. M_x = mean under Red Bull condition; M_w = mean under water condition. All values are means (SD). p-values are two tailed. *p < .05.

Summary of Results

Neither the Word Span Backward nor the 2-back task yielded statistically significant differences between Red Bull and water conditions. The hypothesis that Red Bull would produce modest improvements in working memory relative to water was not supported. However, the effect size for the WSB task ($d = 0.48$) is consistent with a small-to-medium practical effect, and the null result may partly reflect insufficient statistical power given the small sample size.

Discussion

The primary aim of this study was to determine whether consuming a 250-milliliter Red Bull energy drink produces measurable improvements in working memory performance relative to plain water in adolescents. Neither the Word Span Backward nor the 2-back task revealed statistically significant differences, and the hypothesis was not supported. These findings are discussed in light of the existing literature, methodological considerations, and their implications for adolescent health.

Comparison with Prior Research

The null results are consistent with a portion of the adult literature. Moore (2014) similarly found no significant effects of Red Bull on reaction time or executive functioning. Eckerson et al. (2013) reported null effects of physical performance domains. The present findings extend this null pattern to the adolescent working memory domain.

However, the findings contrast with positive adult studies. Wesnes et al. (2017) reported significant improvements in working memory and reaction time following 250 ml of Red Bull in university students. One explanation for the discrepancy is developmental: adolescents may experience attenuated cognitive benefits due to differences in caffeine metabolism and receptor sensitivity (Temple, 2009). Alternatively, the doses of active ingredients in a single 250 ml may be insufficient to produce effects large enough to detect with this sample size.

The trend favouring Red Bull on the WSB task ($d = 0.48$) is noteworthy. Although not significant, this effect size is consistent with the modest improvements predicted by the hypothesis and with findings from ingredient-specific research suggesting glucose-mediated memory benefits (Giles et al., 2012). The absence of a trend on the 2-back task may reflect the greater cognitive demands of that measure, which may require a higher level of stimulant-induced arousal to produce measurable effects, or may simply reflect floor or ceiling effects within the sample.

Role of Hydration

The finding that water produced comparable performance to Red Bull is consistent with Khan et al. (2019), who demonstrated that hydration alone can improve outcomes in cognition in children. The mechanism underlying water's cognitive benefits likely involves increased cerebral

blood flow and optimised electrolyte balance, which supports neural transmission efficiency. When either hydration state is even mildly suboptimal, working memory performance degrades; consuming any fluid, stimulant-containing or not, may therefore partially correct its deficit and produce overlapping performance outcomes. This creates a ceiling effect in which Red Bull's stimulant components cannot demonstrate additional benefit above the baseline hydration correction already provided by both beverages. Because both beverages provided equivalent hydration, it is possible that the hydration component of Red Bull contributed to working memory performance, partially masking any additional stimulant effects. Future research should include a no-beverage control condition, which would allow researchers to quantify the isolated contribution of hydration and estimate whether Red Bull's stimulant ingredients produce any benefit above and beyond simply consuming fluids. Theoretically, a no-beverage condition would likely yield lower performance than either energy drink, helping clarify the hydration effect from the stimulant effect.

Individual Variability

The high standard deviations observed, particularly in the water condition for 2-back accuracy ($SD = 24.88$), suggest substantial individual variability in baseline cognitive performance and beverage responses. Some participants showed clear Red Bull advantages (e.g., Participant 3: WSB Red Bull = 7, Water = 4), while others showed equivalent or reversed patterns. This variability is consistent with literature suggesting that caffeine's effects on cognition are moderated by individual differences such as habitual caffeine use, genetic polymorphisms in caffeine metabolism, and baseline arousal level (Temple, 2009). With only eight participants, individual differences exerted disproportionate influence on group means and variance, which reduced statistical power.

Limitations

Several limitations must be acknowledged when interpreting the findings of this study.

The most significant limitation is the small sample size. The original target was 16-20 participants; however, only eight completed the study due to two compounding recruitment challenges, despite having 20 initial signups. First, several potential participants failed to return complete parental consent forms despite multiple reminders, which is a common logistical barrier in research that involves minors. Second, scheduling conflicts prevented a number of eligible participants from attending both required sessions within the data collection window. The resulting sample of eight participants greatly reduced statistical power, which made it difficult to detect small-to-medium effects. A post-hoc power analysis using the obtained WSB effect size indicated that a substantially larger sample would be needed to achieve adequate power. The current study was therefore underpowered, and the null findings should be interpreted with this in mind rather than as strong evidence of no effect.

A second limitation is the single-blind rather than the double-blind design. Although the researcher was blinded to beverage assignment, participants were fully aware of which beverage they were consuming due to the distinct taste and appearance of Red Bull versus water. This created the potential for expectancy effects, whereby participants who believed they consumed Red Bull may have performed differently due to that belief rather than due to the beverage's pharmacological properties. Critically, the expectancy bias could operate in different directions depending on beverage order. Participants in Order B, who received water first and knew they would receive Red Bull in Session 2, may have approached their second session with anticipation and motivation: a form of anticipatory arousal unrelated to the beverage itself. While counterbalancing distributes order effects across groups, it does not eliminate within-participant

expectancy differences. Future research should use a flavoured placebo closely matched to Red Bull in taste, colour, and carbonation to neutralise this asymmetric expectancy confound.

Third, the absence of a no-beverage baseline condition limits the ability to determine whether working memory performance under either beverage condition differs from an untreated baseline. Including a third condition in future crossover designs would allow for stronger causal conclusions about the direction of any beverage effects.

Fourth, the homogeneity of the sample, adolescents aged 13-17 recruited primarily through social networks, limits the generalisability of the findings. The sample may over-represent individuals with higher socioeconomic status or particular demographics. Additionally, the relative narrow age range and absence of sex-based analysis preclude conclusions about how age, sex, or other demographic factors moderate the observed effects. This is a meaningful omission given that differences in gender and their effects on caffeine metabolism are well documented; females, for instance, tend to metabolise caffeine more slowly than males do due to many reasons, including hormonal interactions (Dillon et al., 2019; Temple, 2009), which could produce systematically different cognitive responses to the same dose. Future studies should include sufficient sample sizes to stratify analyses by sex and age subgroup.

Finally, the post-session side effects questionnaire data were not formally analysed in the current study due to sample size constraints. Subjective reports of alertness, mood, and perceived cognitive performance could have provided useful contextual information and should be incorporated into future analyses. Additionally, a minor procedural variation occurred whereby pre- and post-session questionnaires were administered orally rather than in written form from Participant 4 and onward for both sessions, which may have introduced minor social desirability effects in participant responses.

Conclusion

This study investigated whether a 250-milliliter serving of Red Bull energy drink produces greater improvements in working memory performance than plain water in adolescents aged 13-17. Using a randomised, single-blind, within-subjects crossover design, no statistically significant differences were observed on either the Word Span Backward task or the 2-back task. Although a small-to-medium descriptive trend favoured Red Bull on the verbal working memory measure ($d = 0.48$), this did not reach statistical significance and must be interpreted carefully given the study's underpowered sample of eight participants.

These findings do not support the hypothesis that Red Bull meaningfully enhances adolescent working memory beyond what water alone provides, at least under the conditions tested. The results are consistent with mixed findings in the adult literature and align with theoretical expectations that adolescents may show weakened or inconsistent cognitive responses to the stimulant ingredients in energy drinks.

Given the widespread and increasing consumption of energy drinks among adolescents, further research in the domain is warranted. Future studies should aim for adequately powered samples of at least 36 participants total in a within-subjects design, employ double-blind designs using flavoured placebos, include a no-beverage control condition, and consider individual difference moderators such as habitual caffeine use, sex, and age.

Implications

The findings of this study carry meaningful implications across several communities of practice, even in the context of null statistical results. For public health practitioners and adolescent healthcare providers, the non-significant outcomes alongside a small-to-medium trend

($d = 0.48$) on the WSB task underscore the importance of caution when advising teenagers about energy drink use. The absence of a demonstrated cognitive benefit means this study found no evidence to support the use of Red Bull by adolescents for the purpose of working memory improvement. Healthcare providers can use this evidence as reinforcement of existing guidance from the AACAP and the American Academy of Pediatrics, which already advise against adolescent energy drink consumption, now supplemented by direct cognitive outcome data from the target population.

For educators and school administrators, these results are directly relevant to ongoing debates about energy drink policies on school campuses. Schools seeking evidence-based justification for restricting energy drink access can point to the lack of demonstrated cognitive benefit alongside well-documented physiological risks such as cardiac electrophysiological changes (Elitok et al., 2015) and heightened anxiety responses in adolescents (Temple, 2009). The practical implication for students—that Red Bull does not demonstrably improve the working memory needed for tests, homework, or concentration—challenges the marketing-driven belief that fuels adolescent consumption patterns (Malinauskas et al., 2007; Raj et al., 2018).

For the scientific community, this study contributes a rare experimental dataset on adolescent working memory and energy drink consumption using a controlled crossover design. The effect size of $d = 0.48$ on the WSB task, while not statistically significant with the current sample, is large enough to merit replication with an adequately powered study (approximately 36 participants). Future studies should also address the methodological limitations identified here: double-blind design using flavoured placebos, a no-beverage control arm, sex-based analysis, and formal quantitative analysis of post-session subjective responses. Such responses would

provide substantially stronger evidence base from which to derive adolescent-specific public health guidelines, which are currently absent from both clinical and educational settings. In the end, replacing assumption-based guidance with adolescent-specific data is perhaps the most important contribution this line of research can make.

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Appendix A

Participant Recruitment Flyers

Multiple versions of the recruitment poster were used across different platforms and distribution contexts. Each poster described the study purpose, eligibility criteria (age 13-17, no daily caffeine use, general good health), and time commitment (two sessions) either through directly showing it on the poster, or by providing a link to a page with more information on the study (see Appendix J).

Figure A1



Figure A2



Figure A3



Appendix B

Participant Interest Form

This form collected initial participant contact information, demographic information (age, sex) and caffeine consumption patterns. Used to screen candidates prior to formal consent. Included an information page describing the study in full (see Appendix J). Administered via: <https://tally.so/r/9qGZKp>

Red Bull Research Interest Form

Interest Form - "The Effects of Red Bull on Working Memory in Teenagers"

Thank you for your interest in participating! This short form helps determine if you qualify for the study and ensures that parental consent can be obtained before participation. Please answer honestly. Completing this form does not automatically sign you up! You and **your parents** will receive a parental consent form if eligible. This form should take around 5-7 minutes to complete.

For **more information** on this study and what you will be doing, click here (https://drive.google.com/file/d/1w6cniO7_cRvylAIWr6kvFj1OeCETAwZV/view?usp=sharing).

- Section 1: Participant Information
 - What is your (participant) full name?
 - Participant Age

- Grade in school
- Gender (optional)
- Participant Email Address (the one you check the most)
- Participant Phone Number
- Parent/Guardian Full Name
- Parent/Guardian Phone Number
- Parent/Guardian Email Address
- Section 2: Health and Eligibility
 - Do you have any heart conditions, anxiety disorders, or caffeine sensitivities?
 - Do you regularly consume caffeinated drinks (coffee, soda, or energy drinks)?
 - How often do you consume caffeinated drinks? (Never; Occasionally (1–2 per week); Frequently (3+ per week))
 - Do you have any known allergies or medical conditions that might affect participation? (Type N/A if not applicable)
 - Are you willing to abstain from caffeine for 24 hours before each session? (Yes/No)
 - Are you available for two short testing sessions (each ~40 minutes) at least one week apart? (Yes/No)
- Section 3: Scheduling Preferences
 - What time(s) and day(s) work best for testing?
- Section 4: Acknowledgement
 - By submitting this form, I understand that I am expressing interest in this research study and that my parents/guardian will need to provide consent before I can

participate.

I understand.

Signature

Appendix C

Combined Parental Consent and Participant Assent Form

Required before participation. Parent/guardian provided legal consent; participants provided written assent on the same form. Described study purpose, procedures, duration, risks, and voluntary withdrawal rights. Administered via: <https://tally.so/r/OD1z68>

Red Bull Research Parental Consent and Participant Assent Form

Project Title

Effects of an Energy Drink on Working Memory Performance in Adolescents

Identification of Investigators and Purpose of Study

You are being asked to allow your child (aged 13-17) to participate in a research study conducted by x. The purpose of this study is to examine whether consuming a 250 milliliter Red Bull beverage influences working memory performance in teenagers compared with drinking water.

This study is conducted for a high school research project and will not be used for commercial purposes.

Research Procedures

If you agree for your child to participate, you will sign this form once all questions are answered to your satisfaction.

Each participant will complete two research sessions at least one week apart. Each session will last approximately thirty minutes and will include:

- Baseline questionnaire regarding sleep, alertness, and recent caffeine use
- Consumption of one beverage (Red Bull or water)
- A thirty minute waiting period
- A short post session questionnaire regarding any physical or mental effects

A third party will prepare beverages so the researcher does not know which drink is served during testing. Participants will not be explicitly told the coded beverage condition by the researcher during the session; however, because Red Bull and water differ in taste, participants will be able to infer which beverage they received. The study will take place in a quiet room at a mutually agreed location.

What Participation Involves:

- Two in-person testing sessions (approximately 1 hour each), scheduled at least 7 days apart
- Location: Testing will occur in a quiet, private space. The specific location will be determined based on participant preferences and convenience (e.g., a school, your home, a library study room, or another mutually agreed upon location with stable Wi-Fi and minimal distractions)
- Each session includes: drinking a beverage (either Red Bull or water in an opaque cup), a 30-minute waiting period with quiet activities, and completing two brief memory tests on a computer
- Sessions will be scheduled at the same time of day (between 8 AM - 2 PM) to ensure consistency

Time Required

Participation will require approximately one hour total across two sessions.

Safety & Risks

The investigator perceives minimal risk. Red Bull contains about eighty milligrams of caffeine. The caffeine amount (approximately 80 mg) is below the recommended daily limit for adolescents.

Possible temporary effects include:

- Faster heart rate
- Feeling restless or jittery
- Mild stomach discomfort
- Temporary difficulty sleeping if consumed late in the day

Participants with medical conditions or caffeine sensitivity will not be included in the study. Water will be provided, and participants may stop at any time.

Eligibility Criteria: You may not be eligible if you have:

- Cardiovascular (heart) conditions
- Caffeine sensitivity or allergy
- Anxiety or sleep disorders
- Medications that interact with caffeine
- Irregular caffeine consumption patterns
 - BMI outside healthy ranges for your age

Benefits

There are no guaranteed direct benefits to participants. Students may gain experience in scientific research and cognitive testing. The results may increase understanding of energy drink effects in adolescents.

Confidentiality

All information collected will remain confidential. Participants will be assigned a code number and names will not appear in any report or presentation. Data will be stored securely and accessible only to the researcher. After the project is complete, identifying information will be deleted.

Participation and Withdrawal

Participation is voluntary. You and your child may withdraw at any time without penalty. Participants may decline to answer any question.

Questions about the Study

If you have questions or concerns, contact:

Researcher Name: x

School: x

Email: x

Questions about Your Rights as a Research Subject

x

x

x

x

Giving of Consent

I have read this consent form and understand what is requested. I freely consent to my child's participation.

Parent/Guardian Name: _____

Parent/Guardian Signature: _____

Today's Date: _____

Participant Name: _____

Participant Signature: _____

Today's Date: _____

Appendix D

Pre-Session Health Screening and Baseline Questionnaire Template

Administered at the start of each session (both Session 1 and Session 2). Confirmed participant compliance with pre-session requirements (24-hour caffeine abstinence, 7-9 hours of sleep, light meal, no strenuous exercise). Also collected baseline alertness and mood ratings to establish within-session cognitive state prior to beverage consumption. Note: For Participants ID 1, 2, and 3, this questionnaire was administered in written form via the document below. From Participant 4 and onward, questions were administered orally by the researcher using this document as a guide, with responses recorded verbatim in session field notes.

Pre Session Health Screening and Baseline Questionnaire

Study Title: Effects of a Single Energy Drink on Working Memory Performance in Adolescents

Participant Information

Name: _____

Date: _____

Session Number: ☐ Session One ☐ Session Two

Age: _____ Grade: _____ Sex: ☐ Male ☐ Female ☐ Prefer not to say

Section A: Medical and Safety Screening

(Answer each item honestly. This information ensures safe participation.)

1. Has a healthcare professional ever informed you of heart conditions, blood pressure issues, fainting, seizures, or other medical conditions that may be affected by caffeine?
☐ Yes ☐ No
2. Do you take any medications that caution against caffeine?
☐ Yes ☐ No
3. Do you have any known caffeine sensitivity or history of adverse reactions to caffeine?
☐ Yes ☐ No

If yes, describe briefly: _____

4. Do you have any food or ingredient allergies?

☐ Yes ☐ No

If yes, list here: _____ If you answered yes to

any item above, the researcher will review before participation continues.

Section B: Habitual Caffeine Intake

How often do you normally consume caffeinated drinks?

- ☐ Never
- ☐ Less than once per week
- ☐ One to three times per week
- ☐ Four or more times per week
- ☐ Daily

Approximate caffeine consumed on a typical day:

- ☐ None
- ☐ One drink
- ☐ Two drinks
- ☐ Three or more drinks

Section C: Pre Session Compliance Checklist

Confirm the following instructions were followed in the last twenty four hours: ☐ No caffeine consumed

- ☐ No energy drinks, soda, coffee, or tea
- ☐ No strenuous exercise
- ☐ Slept at least seven to nine hours last night

Hours slept: _____

- ☐ Ate a normal meal at least one hour before this session

If any item is not confirmed, explain: _____

Section D: Baseline Cognitive and Physical State

Circle one number for each (1 = very low, 5 = very high)

Alertness: 1 2 3 4 5

Focus: 1 2 3 4 5

Mood: 1 2 3 4 5

Stress: 1 2 3 4 5

Physical energy: 1 2 3 4 5

Current symptoms (check all that apply):

☐ None ☐ Headache ☐ Nausea ☐ Dizziness ☐ Upset stomach ☐ Fatigue ☐

Other: _____

Participant Declaration

I confirm that my answers are accurate and that I followed the required instructions for this research session.

Participant Signature: _____ Date: _____

Researcher Review

Participant eligible and cleared to continue? ☐ Yes ☐ No

Notes (if applicable): _____

Researcher Signature: _____ Date: _____

Appendix E

Word Span Backward (WSB) Test — Administrator Guide

Standardised verbal working memory assessment. Administered by the researcher approximately 35 minutes post-beverage consumption (at expected peak caffeine absorption). Participants heard sequences of words and repeated them in reverse order; sequences increased in length. Scores reflect the number of sequences accurately recalled.

Word Span Backward Test - Manual Administration Guide

Materials Needed

- This administration guide
- Recording sheet (see below)
- Pen/pencil
- Quiet testing environment
- Timer/stopwatch

Instructions for Administrator

Before Testing

1. Fill in participant information on the recording sheet
2. Ensure quiet environment with minimal distractions
3. Sit across from participant so they cannot see your materials

Script to Read to Participant

"In this test, I will read you a list of words. After I finish reading all the words, I want you to repeat them back to me in REVERSE ORDER - that means backwards, starting with the last word I said.

For example, if I say: CAT - DOG - TREE, you would say: TREE - DOG - CAT The

lists will get longer as we go. Just do your best. Do you have any questions?"

[Answer any questions, then proceed]

"Are you ready? Here is the first list."

Administration Guidelines

Timing:

- Read each word clearly at a rate of approximately **1 word per second** •
- Pause for **1 second** between words
- After the last word, wait for the participant to respond
- Do not repeat words unless there was a clear disruption (external noise, etc.)

Recording Responses:

- Write down EXACTLY what the participant says in the order they say it •
- Mark as correct (✓) or incorrect (✗)
- A response is CORRECT only if all words are recalled in the exact reverse order

Discontinuation Rule:

- Continue through all trials regardless of performance
- Do NOT tell the participant if they are correct or incorrect during testing

Test Trials

SPAN 2 (3 trials)

Trial 1:

Read: BOOK - CHAIR

Correct answer: CHAIR - BOOK

Participant response: _____

Correct? ☐ Yes ☐ No

Trial 2:

Read: LAMP - DOOR

Correct answer: DOOR - LAMP

Participant response: _____

Correct? ☐ Yes ☐ No

Trial 3:

Read: DESK - PHONE

Correct answer: PHONE - DESK

Participant response: _____

Correct? ☐ Yes ☐ No**SPAN 3 (3 trials)****Trial 4:**

Read: CAR - TREE - FISH

Correct answer: FISH - TREE - CAR

Participant response: _____

Correct? ☐ Yes ☐ No**Trial 5:**

Read: BALL - CLOUD - RING

Correct answer: RING - CLOUD - BALL

Participant response: _____

Correct? ☐ Yes ☐ No**Trial 6:**

Read: FORK - SHOE - BIRD

Correct answer: BIRD - SHOE - FORK

Participant response: _____

Correct? ☐ Yes ☐ No**SPAN 4 (3 trials)****Trial 7:**

Read: HOUSE - TIGER - CLOCK - BREAD Correct
 answer: BREAD - CLOCK - TIGER - HOUSE Participant
 response: _____ Correct? ☐ Yes ☐ No

Trial 8:

Read: STONE - RIVER - APPLE - BRUSH Correct
 answer: BRUSH - APPLE - RIVER - STONE Participant
 response: _____ Correct? ☐ Yes ☐ No

Trial 9:

Read: PAPER - MUSIC - PLANT - LIGHT Correct
 answer: LIGHT - PLANT - MUSIC - PAPER Participant
 response: _____ Correct? ☐ Yes ☐ No

SPAN 5 (3 trials)**Trial 10:**

Read: SHIRT - QUEEN - BEACH - HORSE - KNIFE

Correct answer: KNIFE - HORSE - BEACH - QUEEN - SHIRT

Participant response: _____

Correct? ☐ Yes ☐ No**Trial 11:**

Read: GRASS - MONEY - THUMB - SOUTH - FRAME

Correct answer: FRAME - SOUTH - THUMB - MONEY - GRASS

Participant response: _____

Correct? ☐ Yes ☐ No**Trial 12:**

Read: BREAD - SMILE - WATCH - NIGHT - CLOUD

Correct answer: CLOUD - NIGHT - WATCH - SMILE - BREAD

Participant response: _____

Correct? ☐ Yes ☐ No**SPAN 6 (2 trials)****Trial 13:**Read: FINGER - WINDOW - GARDEN - BOTTLE - SQUARE - WINTER Correct
answer: WINTER - SQUARE - BOTTLE - GARDEN - WINDOW - FINGER Participant
response: _____Correct? ☐ Yes ☐ No**Trial 14:**Read: ORANGE - BUTTON - VALLEY - PURPLE - TEMPLE - CIRCLE Correct
answer: CIRCLE - TEMPLE - PURPLE - VALLEY - BUTTON - ORANGE Participant
response: _____Correct? ☐ Yes ☐ No**Scoring****Calculate Total Score**

- Total Correct: _____ / 14
- Percentage: _____ %

Determine Maximum Span Achieved

A span level is "achieved" if the participant gets at least 2 out of 3 trials correct at that span length (or 1 out of 2 for Span 6).

- Span 2: _____ / 3 correct → Achieved? ☐ Yes ☐ No
- Span 3: _____ / 3 correct → Achieved? ☐ Yes ☐ No
- Span 4: _____ / 3 correct → Achieved? ☐ Yes ☐ No
- Span 5: _____ / 3 correct → Achieved? ☐ Yes ☐ No
- Span 6: _____ / 2 correct → Achieved? ☐ Yes ☐ No

Maximum Span Achieved: _____

Recording Sheet Template

Participant ID: _____

Session: ☐ Session 1 ☐ Session 2

Date: _____

Time: _____

Administrator: _____

Testing Environment Notes:

- Location: _____
- Any interruptions or concerns: _____

After Testing

Thank the participant:

"Thank you for participating. You did a great job!"

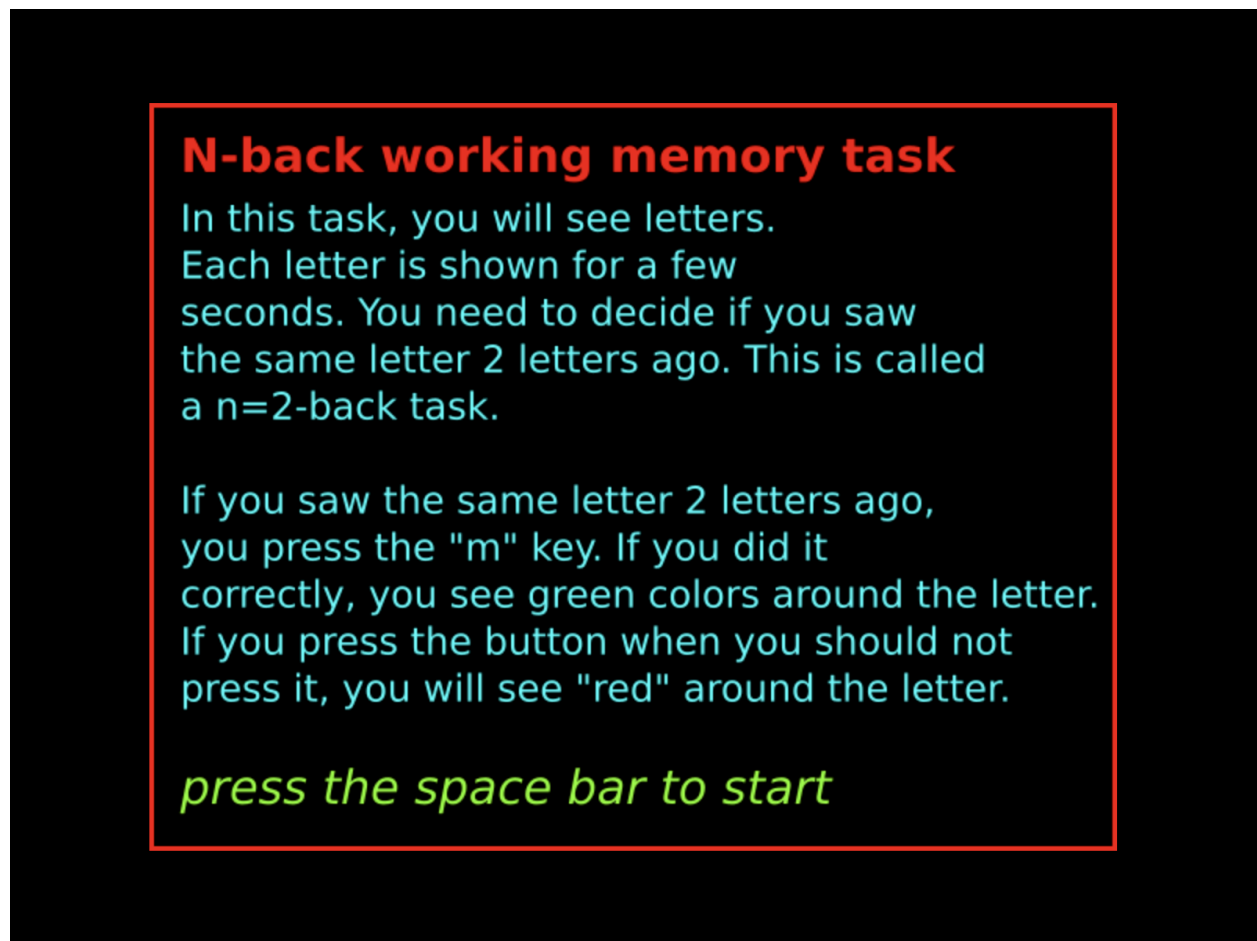
Do NOT discuss their performance or give feedback about correctness.

Appendix F

2-Back Task — PsyToolKit Setup

Computerised spatial working memory assessment measuring updating ability. Administered immediately following the WSB task during the same peak absorption window. Participants viewed a sequential stream of letters and indicated when the current letter matched the one presented two trials prior. Accuracy (%) was recorded as the primary dependent variable.

1. The task instructions:

A screenshot of a black screen with a red rectangular border containing white and green text. The text provides instructions for an N-back working memory task, explaining the goal (deciding if a letter seen 2 trials ago is repeated) and the response (pressing the 'm' key). It also indicates that correct responses are highlighted in green and incorrect ones in red. The instruction to 'press the space bar to start' is written in green at the bottom.

N-back working memory task

In this task, you will see letters.
Each letter is shown for a few seconds. You need to decide if you saw the same letter 2 letters ago. This is called a n=2-back task.

If you saw the same letter 2 letters ago, you press the "m" key. If you did it correctly, you see green colors around the letter. If you press the button when you should not press it, you will see "red" around the letter.

press the space bar to start

2. The task parameters

- Platform: PsyToolKit (psytoolkit.org)

- Stimulus type: Letters presented sequentially
- Target condition: Current letter matches letter presented 2 trials prior
- Trial count: 25 trials per block (with 2 real blocks and 1 training block)
- Response keys: M
- Accuracy recorded as: Percentage of correctly matched items; the task also records missed items, false alarms, and overall trial score

Appendix G

Post-Session Side Effects and Cognitive State Questionnaire Template

Administered at the conclusion of each session. Captured self-reported side effects (e.g., increased heart rate, jitteriness, headache) and subjective perceptions of alertness and cognitive performance. Also included a beverage identification question to assess blinding integrity. Note: For Participants ID 1, 2, and 3, this questionnaire was administered in written form via the document below. From Participant 4 and onward, questions were administered orally by the researcher using this document as a guide, with responses recorded verbatim in session field notes.

Post Session Side Effects and Cognitive State Questionnaire

Study Title: Effects of a Single Energy Drink on Working Memory Performance in Adolescents

Participant Name: _____

Date: _____

Session Number: ☐ Session One ☐ Session Two

Section A: Physical Sensations

Rate each from 1 (none) to 5 (strong)

Headache: 1 2 3 4 5

Stomach discomfort: 1 2 3 4 5

Jitteriness or restlessness: 1 2 3 4 5

Nausea: 1 2 3 4 5

Change in heart sensation: 1 2 3 4 5

Any other physical sensations experienced:

Section B: Cognitive State

Rate each from 1 (very low) to 5 (very high)

Sleepiness: 1 2 3 4 5

Focus: 1 2 3 4 5

Energy level: 1 2 3 4 5

Mood: 1 2 3 4 5

Any other mental or emotional changes noticed:

Section C: Perceived Task Performance

How well do you think you performed on the memory tasks?

1 2 3 4 5 (1 = very poorly, 5 = very well)

Section D: Blinding Check

Which drink do you think you received during this session?

- ☐ Water
- ☐ Energy drink
- ☐ Not sure

How confident are you in your guess?

1 2 3 4 5 (1 = not confident, 5 = very confident)

Section E: Additional Comments

Please describe anything unusual you experienced or noticed during testing.

Participant Declaration

I confirm that the information above is accurate to the best of my knowledge.

Participant Signature: _____ Date: _____

Researcher Signature: _____ Date: _____

Appendix H

Session Field Notes Template

Used by the researcher during both sessions to document participant behavior, protocol compliance, beverage identification attempts, and any unexpected errors. Provided qualitative context for interpreting quantitative outcomes.

Session Field Notes

Study Title: Effects of Red Bull Energy Drink vs. Water on Working Memory Performance in Teenagers

Session Information

Participant ID: _____
Session Number: ☐ Session 1 ☐ Session 2
Date: _____
Session Start Time: _____
Session End Time: _____
Researcher Initials: _____

Pre-Session Compliance Verification

Pre-Session Health Screening Completed: ☐ Yes ☐ No
All eligibility criteria confirmed: ☐ Yes ☐ No

Compliance Checklist (from participant self-report):

- **Caffeine abstinence (24 hours):** ☐ Yes ☐ No
 - If no, details: _____
- **Sleep (7-9 hours):** ☐ Yes ☐ No
 - Hours reported: _____
- **Light breakfast (≥1 hour prior):** ☐ Yes ☐ No
 - Time of breakfast: _____

- No strenuous exercise today: ☐ Yes ☐ No

○ If no, details: _____

Non-compliance notes (if any):

Beverage Administration

Beverage Code: _____ (e.g., Cup A, Cup B)

Beverage Start Time: _____

Beverage End Time: _____

Total Consumption Time: _____ minutes

Full beverage consumed: ☐ Yes ☐ No

If no, amount consumed: _____ ml

Observations during consumption:

Waiting Period (30 minutes)

Waiting Period Start Time: _____

Waiting Period End Time: _____

Activities during waiting period: ☐ Reading ☐ Coloring ☐ Phone use ☐ Other:

Participant demeanor/behavior:

Any reported discomfort or concerns:

Cognitive Testing Period

Testing Start Time (approx. 35 min post-consumption): _____

Word Span Backward Test (5 minutes)

Test Start Time: _____

Test End Time: _____

Test completed successfully: ☐ Yes ☐ No

Observations:

- Participant engagement: ☐ Highly engaged ☐ Moderately engaged ☐ Low engagement
- Participant appeared: ☐ Focused ☐ Distracted ☐ Tired ☐ Alert •
- Technical issues: ☐ None ☐ Yes (describe): _____ •
- Behavioral notes: _____

2-Back Task (10 minutes)

Test Start Time: _____

Test End Time: _____

Test completed successfully: ☐ Yes ☐ No

Observations:

- Participant engagement: ☐ Highly engaged ☐ Moderately engaged ☐ Low engagement
- Participant appeared: ☐ Focused ☐ Distracted ☐ Tired ☐ Alert •
- Technical issues: ☐ None ☐ Yes (describe): _____ •
- Behavioral notes: _____

Post-Test Observations

Post-Test Questionnaire Completed: ☐ Yes ☐ No

Time Completed: _____

Visible side effects observed (check all that apply):

- ☐ None observed
- ☐ Increased restlessness/fidgeting
- ☐ Elevated energy level
- ☐ Fatigue
- ☐ Other: _____

Participant comments or feedback:

Environmental Conditions

Testing environment: ☐ Quiet ☐ Some noise ☐ Distracting

Lighting: ☐ Adequate ☐ Too bright ☐ Too dim

Temperature: ☐ Comfortable ☐ Too warm ☐ Too cold

Interruptions during session: ☐ None ☐ Yes (describe): _____

Data Quality Notes

Data integrity concerns: ☐ None ☐ Yes

If yes, describe:

Reasons to exclude data from analysis (if any):

Additional Notes

Any unusual circumstances, observations, or concerns:

Session Completion

Total Session Duration: _____ minutes

Session completed without issues: ☐ Yes ☐ No

Participant thanked and dismissed: ☐ Yes

Next session scheduled (if applicable): ☐ Yes ☐ No ☐ N/A

Next session date/time: _____

Researcher Signature: _____ Date: _____

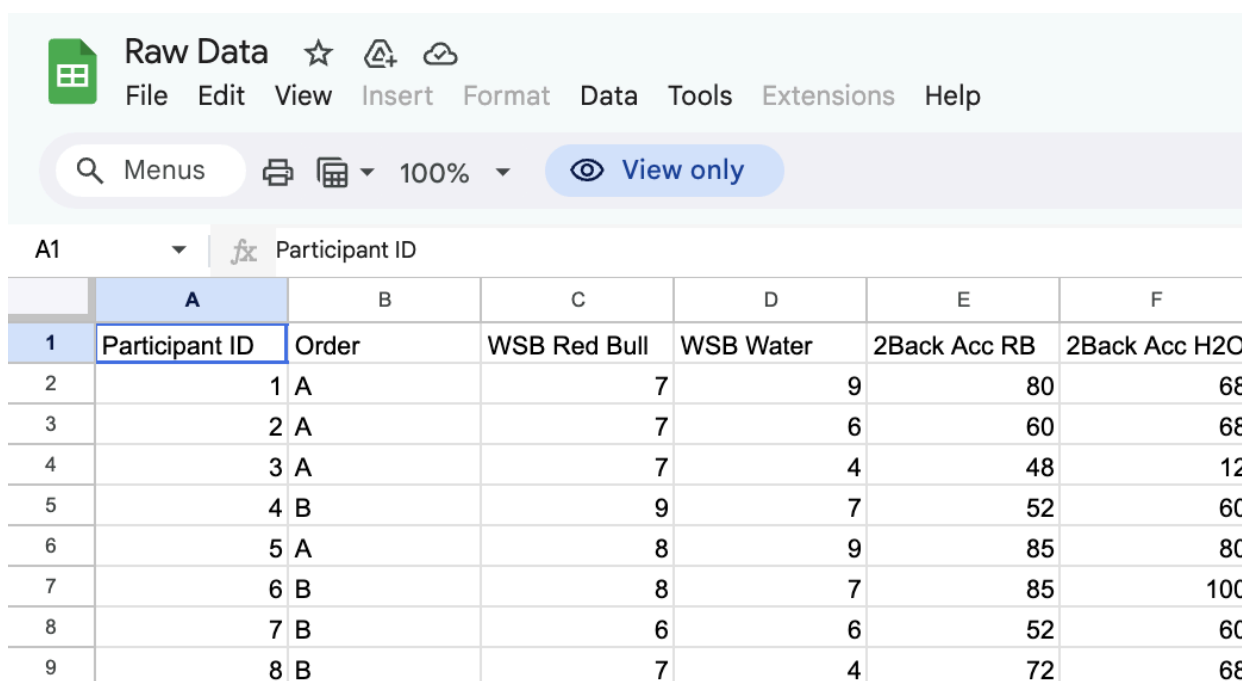
For Researcher Use After Session:

Data entered into spreadsheet: ☐ Yes ☐ No **Date:** _____ **Field**
notes reviewed for completeness: ☐ Yes **Date:** _____

Appendix I

Raw Data Spreadsheet

Complete anonymised dataset containing WSB and 2-back accuracy scores for all eight participants under both beverage conditions, coded by participant ID. Serves as the primary dataset used for all statistical analyses reported in the Results section.



	A	B	C	D	E	F
1	Participant ID	Order	WSB Red Bull	WSB Water	2Back Acc RB	2Back Acc H2O
2	1	A	7	9	80	68
3	2	A	7	6	60	68
4	3	A	7	4	48	12
5	4	B	9	7	52	60
6	5	A	8	9	85	80
7	6	B	8	7	85	100
8	7	B	6	6	52	60
9	8	B	7	4	72	68

Appendix J

Important Information for Research Participation

Google Document linked to the interest form and the flyers: A1, A2, and A3. This explained the purpose of the study, eligibility criteria, requirements before sessions, safety and privacy, outlined the time commitment, and provided contact information for additional inquiry.

Important Information for Research Participation

Study Overview: This research study investigates how consuming a 250 ml Red Bull energy drink compared to plain water affects working memory (the ability to hold and use information for tasks like problem-solving) in teenagers aged 13-17.

What Participation Involves:

- **Two in-person testing sessions** (approximately 1 hour each), scheduled at least 7 days apart
- **Location:** Testing will occur in a quiet, private space. The specific location will be determined based on participant preferences and convenience (e.g., a school, your home, a library study room, or another mutually agreed upon location with stable Wi-Fi and minimal distractions)
- Each session includes: drinking a beverage (either Red Bull or water in an opaque cup), a 30-minute waiting period with quiet activities, and completing two brief memory tests on a computer
- Sessions will be scheduled at the same time of day (between 8 AM - 2 PM) to ensure consistency

Requirements Before Each Session:

- No caffeine consumption for 24 hours prior (no coffee, tea, energy drinks, or soda)
- 7-9 hours of sleep the night before
- A light breakfast at least 1 hour before testing
- No strenuous exercise on testing days

Eligibility Criteria: You may **not** be eligible if you have:

- Cardiovascular (heart) conditions
- Caffeine sensitivity or allergy
- Anxiety or sleep disorders
- Medications that interact with caffeine
- Irregular caffeine consumption patterns

- BMI outside healthy ranges for your age

Safety & Privacy:

- The caffeine amount (approximately 80 mg) is below the recommended daily limit for adolescents
- All information will be kept confidential and anonymous
- You may withdraw at any time without penalty
- Both you and your parent/guardian must provide consent before participation

Time Commitment: Approximately 2-3 hours total (two 1-hour sessions plus travel time)

Questions or Concerns: If you have any questions, want to revoke your interest, or have any other inquiries or concerns, please contact: x

